

Supplementary Material: Diagnosing and Surgically Correcting Conservative Decision Bias in Imitation-Learned Air Combat Policies

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Submission Id: 000

ABSTRACT

Supplementary material for the paper above. Section E records the provenance of every number quoted in the main text; nothing here is required in order to follow the argument.

KEYWORDS

reproducibility, provenance

ACM Reference Format:

Anonymous Author(s). 2027. Supplementary Material: Diagnosing and Surgically Correcting Conservative Decision Bias in Imitation-Learned Air Combat Policies. In *Proc. of the 26th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2027)*, , May 2027, IFAA-MAS, 4 pages.

1 APPENDIX B. THE ENVIRONMENT'S LAUNCH MASK VERSUS THE EXPERT'S EXTRA GATE

The mask. The flat action mask is `[speed(3), heading(5), altitude(1), fire(2)]`. The launch branch is flat index `10.mask[10] = 1` requires *all* of:

The expert's gate. The rule expert never fires merely because `mask[10] = 1`. It additionally requires its own `fire_desired`, a pure function of the observation:

Why "strict subset" is exact, not rhetorical. With `r_max(AA) = 3000 + 5000 · (AA/180)`:

- lower bound: $1500 + 0.25 · (r_{\max} - 1500) > 1500$ whenever $r_{\max} > 1500$, i.e. for every $AA > 0$;
- upper bound: $1500 + 0.75 · (r_{\max} - 1500) = 0.75 · r_{\max} + 375 < r_{\max}$ whenever $r_{\max} > 1500$.

So on every axis the expert's admissible set is a strict subset of the environment's. This is the mechanical content of the paper's central claim that *the expert carries a conservative engagement preference that the environment does not require* — not a claim about the expert's competence.

Consequence to keep explicit: the mask, hence CLR's denominator, is policy-dependent. Firing triggers C1b, which *closes* the window; abstaining leaves it open. A policy that fires therefore collapses its own opportunity set, while a policy that abstains inflates it. Measured on the Figure-2 seed: **41** legal steps for BC versus **4** for SPC (a factor of 10.25; the aggregate in §4.2 is $\approx 10.8\times$). CLR is consequently a **descriptive** statistic across policies; the causal claim rests on the frozen-trajectory enumeration of §4.3, where the opportunity set is held fixed by construction.

2 APPENDIX C. THE ENUMERATED FIRE-POLICY FAMILY

All seven arms are evaluated on the *same frozen maneuver trajectory* per seed (the BC heading/speed actions), so only the launch decision varies (`scripts/fire_oracle_audit.py`):

`dlz_mid` and `dlz_deep` deliberately bracket the expert's own band `[0.25, 0.75]`, so the family contains a faithful approximation of the expert's gate as well as the mask-permissive extreme.

3 APPENDIX D. REPRODUCTION COMMANDS

All commands are for the **WSL Ubuntu shell**, from the repository root, using the project interpreter `/home/sean/miniconda3/envs/marl_env/bin/python`. Trace JSONs, logs and `.pth` checkpoints are excluded from version control, except the `E*.json` and `mechanism_*` artifacts, which are tracked.

Primary endpoints (§4.4, §4.5b, Table 4):

Frozen-trajectory enumeration (§4.3, Table 3):

Matched BC-versus-expert pair (§4.4, fresh env per episode — note this path also computes CLR as of 2026-09-17):

The ϵ -difficulty arms are the same two commands with `-difficulty 0.3`. The CLR measurement is the same command again with `--model-id paired_bc_vs_expert_d0_clr` and `--out results/shoot_eval/E3_paired_bc_vs_expert_d0_n400_s20000_v3_clr.json`.

Figures 1–3 and the auto-generated matrix:

`make_paper_figures.py` reads every plotted value from the tracked `E*.json` artifacts and holds **no numeric constants**, so Figures 1 and 3 cannot drift away from the text; if an artifact is missing the figure renders pending and warns rather than substituting a value from another basis. `audit_paper_numbers.py` recomputes the headline numbers straight from the artifacts and then checks that the draft still quotes them, including two must-be-absent guards: the abstract's expert-CLR placeholder (once the CLR run completes) and the bogus $2.5e-65$ p-value produced by feeding tie counts into `exact_mcnemar`. Both guards are themselves documented in the draft, so the checker ignores occurrences sitting inside a line that is explicitly talking about the trap.

`collect_matrix.py` reads **only** each file's `run_meta` — never filenames — and skips any artifact whose name is marked `EXCLUDED_`, so retired evidence cannot silently re-enter the table.

4 APPENDIX E. REPRODUCIBILITY AND ARTIFACT PROVENANCE

Supporting evidence for claims made in the main text. The reasoning each item supports is stated in the main text; what follows is the raw material needed to re-run or re-check it.

#	condition	constant (src/environment/singlecombat_shoot_task.py)
C1	missiles remaining > 0	NUM_MISSILES = 4
C1b	steps since last launch $\geq$ MIN_ATTACK_INTERVAL	30 steps = 6 s at 5 Hz
C2	target alive	—
C3	ATA < MAX_ATTACK_ANGLE	15.0 deg
C4	MIN_ATTACK_DISTANCE $\leq$ range $\leq$ 3000 + 5000 · (AA/180)	1500.0 m
C5	closure < 0 (closing on the target)	—

axis	environment permits	expert additionally requires	relation
ATA	< 15.0 deg	< ATA_STRICT = 10.0 deg	strictly tighter
closure	< 0 m/s	< -CLOSURE_MIN = -5.0 m/s	strictly tighter
range	1500 m $\leq$ r_max(AA)	DLZ depth in [0.25, 0.75], where depth = (range - 1500)/(r_max - 1500)	strictly tighter and two-sided

arm	rule
asap	fire at the first legal step (the cooldown is already inside the mask), then whenever legal
delay_30	fire at the first legal step $\geq$ first-legal + 30
delay_60	fire at the first legal step $\geq$ first-legal + 60
dlz_mid	fire only when DLZ depth $\in$ [0.4, 0.6]
dlz_deep	fire only when DLZ depth $\geq$ 0.6
interval_100	fire ASAP, then at most every 100 steps while legal
bc	the inherited BC launch head (the reference arm)

4.1 E.1 The environment-lifecycle confound — affected scripts and artifacts

\S4.4 states the confound and why the reused-env numbers are excluded. The specifics: both paths import the same rule functions (hdg_label / spd_label / fire_desired) at CMD_SPEED = 280, and BaseEnv.__init__ already installs SafetyInterceptor(PIDFlightController()), so neither the rule nor the controller explains the disagreement. The reused-env path is generate_shoot_rule_expert.py --validate, which builds **one** BaseEnv and reuses it for every episode; the fresh-env path is used by eval_paired_bc_vs_expert.py, eval_bc_1v1.py, fire_oracle_audit.py and eval_asap_baseline.py, each of which builds a new environment per episode. Affected artifacts are renamed EXCLUDED_reused_env_expert_*.json, and collect_matrix.py skips anything marked EXCLUDED_. Gate: tests/check_expert_path_consistency.sh.

4.2 E.2 Corroborating CLR: timestamps and the gate on disallowed steps

\S3.2 quotes the two corroborating values (expert 5.96%, BC 7.37%) and their geometry caveat; the exact provenance is:

- data/expert/shoot_rule_expert.npz, 200 episodes, **mtime 2026-09-13 20:22** — this *predates* commit fb48155 (2026-09-14 09:06), which changed the heading-bias range from $U(30, 60)$ to $U(0, 60)$. Its 5.96% is therefore an **old-geometry** measurement.
- results/health_check/fire_hesitancy.json, 200 rollouts, generated **2026-09-16**, i.e. after the change — so the 7.37% beside it is $U(0, 60)$. The two are **geometry-mixed** and are never averaged or presented as a matched pair.

4.3 E.3 Scope reading for the CLR claim

Neither CLR figure alone proves suboptimality: a low CLR diagnoses restriction only, and whether that restriction is *costly* is established separately (\S4.3, \S4.4). The instrumentation and stability cross-checks are in \S4.2.

4.4 E.4 Compute environment (moved from \S4.1)

Single RTX 3060 Laptop GPU (6 GB, cu126, driver 528.79). Closed-loop evaluation: BC $\approx$ 6.7–13.0 s/episode, SPC $\approx$ 10 s/episode, so one $n=400$ arm pair is $\approx$ 1.7 h; rule-expert rollout $\approx$ 6.3 s/episode. All runs execute in WSL2 Ubuntu 22.04 with the project interpreter (/home/sean/miniconda3/envs/marl_env/bin/python); reproduction commands are in Appendix D.

4.5 E.5 Ablation A5 — full training and screening detail (moved from \S4.7)

- **Implementation.** scripts/distill_fire_asap.py --random-fire-head: SPC's freeze pattern is kept exactly (encoder and every maneuver head frozen at θ_{BC}) and only the fire head is discarded and rebuilt from nn.Linear.reset_parameters() — the constructor's own initialisation, so it is a genuine random init (weight std 0.049, bias std 0.044) rather than a hand-rolled approximation of one. Mutually exclusive with --full-network (A5 and A6 are different ablations).
- **Training.** 300 rollout episodes (392091 transitions, 11976 allowed steps; train/val split 9578/2398 allowed steps). The random fire head reaches fire-CE 0.0413 with train recall 100.00% and validation recall 100.00% **in its first epoch** (of 15) and triggers the $\geq$ 99.5% early stop. Identity gates: hdg/spd logits diff = 0.00e+00 — *exactly* zero, not

Quantity	Value	Basis
Expert fire_desired on disallowed steps	4.03% (10336 / 256357) — the gate would fire outside the window too; the mask (mostly cooldown) holds it back	data/expert/shoot_rule_expert.npz

Identity measurement	SPC (fire head only)	A6 (full network)
heading/speed logit max-diff vs θ_{BC}	< 1e-9	8.51
heading/speed action sequences identical	yes	no
launch-mask agreement	yes	yes
validation allowed-window accuracy	≥ 0.995	0.9967

Arm (400 seeds, d=0)	Kill rate	CLR	vs SPC (paired)
BC (inherited launch policy)	46.50%	7.26%	—
A6 — full-network fine-tune	77.50% (310/400)	99.67% (1508/1513)	-13.25 pp , discordant 67 (60 SPC / 7 A6), exact McNemar p = 1.33e-11
SPC — launch head only	90.75% (363/400)	100.00%	—

merely under tolerance — with identical action sequences and VERDICT: PASS.

- **Primary protocol.** 90.75% (363/400) kills, CLR 99.94% (1560/1561) — the CLR denominator differs from SPC's by exactly one legal step (1561 vs 1560), the only trace of the different initialisation anywhere in the evaluation.
- **60-seed screening** (same seed set as A6b; ± 10 pp small-sample spread applies): A5 **95.0%** (id) / **86.7%** (dist2_3k) at 100% window utilisation, bit-for-bit equal to the mask-permissive rule oracle on the same seeds; BC 40.0% / 15.0%. The screening set is kept because it carries the third arm.

4.6 E.6 Ablation A6 — protocol, identity measurements and results (moved from \S4.7)

Protocol. Identical objective, data, loss and initialisation as SPC; the only change is that every parameter is trainable rather than the launch head alone, at a deliberately **more generous** budget (—epochs 40 —lr 3e-4 versus —epochs 15 —lr 1e-2), so that a negative result cannot be attributed to under-training.

Identity measurements — the maneuver is untouched under SPC and demonstrably changed under A6:

Kill rate and CLR at the primary protocol (d=0, U(0,60), the same 400 seeds as Table 4):

Full-network fine-tuning recovers most of the gap (46.50% $\rightarrow$ 77.50%) but remains significantly worse than the localized intervention on identical seeds, and pays for it with an 8.51-logit change to the maneuver.

4.7 E.7 Ablation screening set — the mask-permissive third arm (moved from \S4.7)

The ablation script's own 60-seed evaluation (geometry U(0,60), deterministic masked argmax) runs the id and dist2_3k cells side by side and adds the **mask-permissive rule oracle**, which the 400-seed runs do not:

A6 also loses to a rule that simply fires whenever the mask permits (81.7% vs 95.0%). The set is **60 seeds** and carries roughly ± 10 pp of small-sample spread — the same seeds give BC 40.0% where n=400 gives 46.50% — so these figures are quoted only for the third arm and never alongside the 400-seed numbers as one comparison.

4.8 E.8 Ablation tracking table (project bookkeeping)

4.9 E.9 The E2 rule-oracle comparison — identities and limitations (moved from \S4.4)

eval_asap_baseline.py had no —start-seed, so the oracle could only ever run seeds 0...N-1 and could not be matched to a trained policy's evaluation. Patched, then run on the same seed set:

Identities verified from run_meta: the oracle arm loaded sha256 aad05b45\ldots (BC round1 — the *frozen maneuver* source, correct), SPC loaded 36d79bd9\ldots.

One limitation. eval_asap_baseline.py writes aggregates only, with no per-episode detail, so the identity above follows from construction rather than from a seed-by-seed match. Emitting per-episode records would make it mechanically checkable.

4.10 E.10 The 2026-08-06 frozen-trajectory audit on the old geometry (moved from \S4.3)

Two of the seven arms reproduce exactly (delay_60, dlz_mid) while asap moves from 73.3% to 86.7%. That pattern indicates the *script's* behaviour is unchanged and the benchmark itself became easier by $\approx +4.5$ pp once the heading-bias range was widened (\S4.5). Only the measured column is cited in the paper, always with its geometry attached; the paper's Table 3 carries no geometry-mixed column.

5 ARTIFACT PROVENANCE RECORDS

The records below were collected out of the main text, where they interrupted the argument. Each states which artifact backs which claim.

Arm (60 seeds, same seed set)	id cell	dist2_3k cell	window utilisation (id)
BC (inherited launch policy)	40.0%	15.0%	7.2%
A6 — full-network fine-tune	81.7%	66.7%	98.3%
mask-permissive rule oracle	95.0%	86.7%	100%

#	Ablation	Status	Purpose
A1	Fire-policy oracle enumeration on frozen trajectory (\S4.3)	done (E1 regenerated, tracked)	establishes that the inherited launch policy is sub-optimal, maneuver held fixed
A2	SPC (learned head) vs ASAP rule oracle (\S4.4)	done — exact match on one seed set (90.75% both); completeness check, not independent validation	SPC captures 100% of the oracle gap
A3	difficulty_level = 0.3 (\S4.6)	done — +48.50 pp, 194:0, p=7.97e-59	interactive/reacting opponent
A4	Heading-shift robustness (\S4.5)	done (n=400)	benchmark sensitivity
A5	Fire head from random init, encoder frozen (\S4.7, E.5)	done 2026-09-18 (400-seed primary protocol)	warm-start dependence
A6	Full-network fine-tune on the same objective (\S4.7, E.6)	done 2026-09-18 (400-seed primary protocol)	is freezing merely engineering convenience?
A6b	60-seed screening incl. the mask-permissive arm (E.7)	done, secondary	third-arm comparison only
A7	CLR for other binary decisions	conceptual	generality of the metric (future work)

Arm	Kill	Launches/ep	Hit	lost	Termination reasons
Rule oracle (asap: frozen BC maneuver + fire on every legal step)	363/400 = 90.75%	3.90	1.0000	0	target_killed: 363, timeout: 37
SPC	363/400 = 90.75%	3.90	1.0000	0	target_killed: 363, timeout: 37

Fire policy	Kill rate (measured, U(0,60))	2026-08-06 audit, U(30,60)
asap	86.7%	73.3%
delay_30	38.3%	28.3%
delay_60	10.0%	10.0%
dlz_mid	0.0%	0.0%
dlz_deep	0.0%	3.3%
interval_100	0.0%	0.0%
frozen BC (inherited)	15.0%	6.7%

Citation status (re-checked 2026-09-18 against the live record, not from memory). Verified: Pomerleau (NIPS'88, pp. 305-313; presented 1988, MIT Press 1989 - the year is now confirmed, not inferred), Ross/Gordon/Bagnell (AISTATS 2011), Bojarski (arXiv:1604.07316), Brown et al. (ICML 2019 pp. 783-792), Huo/Wang/Xu (AAAI 2023 pp. 7953-7961), Beliaev/Pedarsani (arXiv:2402.01884), Li Z. et al. (NeurIPS 2023), Parisotto/Ba/Salakhutdinov (ICLR 2016, arXiv:1511.06342), McGrew et al. (JGCD 33(5):1641-1654), Pope et al. (ICUAS 2021 pp. 275-284), Yang et al. (IEEE Access 8:363-378), Li L. et al. (Neurocomputing 584:127591, 2024), Li S. et al. (ACM TAAS 21(1):6, 2026), **Rusu et al. (ICLR 2016 conference track, arXiv:1511.06295), Hinton/Vinyals/Dean (NIPS 2014 Deep Learning Workshop, arXiv:1503.02531 - a workshop paper, so it has no page numbers and none are invented here), LoRA (Hu et al., ICLR 2022, arXiv:2106.09685), ROME (Meng, Bau, Andonian & Belinkov, NeurIPS 2022, Advances in NeurIPS 35, pp. 17359-17372). Nothing on this list is now unconfirmed.**

Provenance. The measured column comes from results/shoot_eval/E1_fire_oracle_dist2_3k_s60.json, which is version-controlled

and carries run_meta with the cell configuration, geometry, difficulty, seed range and the frozen checkpoint's SHA-256. The 2026-08-06 artifact no longer exists; the italic column is all that survived of it, which is why the regenerated file is tracked rather than regenerated on demand.

Artifact status. E7 is **verified against live artifacts** (results/shoot_eval/E7_bc_round1, spc_d0_n400_s20000.json, E7_paired_bc_vs_spc_d0_n400.json).

Artifact status: verified against live artifacts — results/shoot_eval/E5_bc_round1, spc_d03_n400_s20000.json, E5_paired_bc_vs_spc_d03_n400.json, and the two CLR runs E3_paired_bc_vs_expert_d0_n400_s20000_v3, d03_n400_s20000_v2_clr.json. The expert arm is no longer "running": it is the fresh-env paired arm of E3 at both difficulties, and it is what Table 4 reports.